MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, RA, M549 & M549A1
(WITH WARHEAD AND MOTOR BODY)
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-50578C(AR), dated 20 June 1979, and is approved for use by all Departments and Agencies of the Department of Defense.

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Paragraphs 4.5.3.1 and 4.5.3.2: Delete in their entirety and substitute the following:

"4.5.3.1 Functioning. All test projectiles shall be conditioned and fired at 70 degrees Fahrenheit (^{O}F) at Charge 8, M203 series propelling charge conditioned at 70°F. In order to minimize the effects of met conditions the test shall not be fired in either one (for normal inspection) or three (for tightened inspection) series of 10 rounds each. The first two subgroups for tightened inspection shall be fired at a quadrant elevation of 33 degrees and the last subgroups for tightened inspection or the single group for normal inspection shall be fired at a quadrant elevation of 51 degrees. Time lapses of more than 15 minutes between consecutive rounds shall not be permitted and it is desirable that each 10 round series be completed within one (1) hour. Range probable errors shall be calculated for each 10 round subgroup and then a pooled range probable error (for tightened inspection) shall be computed for determination of compliance with 3.6.1.2.1.

$$\text{Subgroup RPE (percent)} = RPEj = \sqrt{\frac{\Sigma X_i - 1/n\ (\Sigma X)^2}{n-1}} \times \frac{100\% \times .6745}{\text{Ave. Range (j)}}$$

FSC: 1320

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Pooled RPE (percent) = RPEp =

$$\sqrt{\frac{RPE_1{}^2(n_1-1) + RPE_2^2\ (n_2-1) + RPE_3^2\ (n_3-1)}{DF}}$$

Where X = Individual horizontal range within each subgroup.

Where DF = Degrees of Freedom = n + n....n(j) - number of subgroups.

4.5.3.2 Safety. The projectiles for this test shall be divided into three equal groups. Group 1 shall be conditioned and fired at -65°F at Charge 8, M203 series propelling charge conditioned at -65°F. Group 2 shall be conditioned and fired at 145°F with an M203 propelling charge conditioned at 145°F. Group 3 shall be conditioned at 145°F and fired with a PXR-6297 proof propelling charge conditioned at 70°F. The cold phase (Group 1) shall be fired at a quadrant elevation of 51 degrees and the hot phases (Group 2 & 3) shall be fired at a quadrant elevation of 24 degrees."

Custodian:
Army - AR

Preparing activity:
Army - AR

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